Placement Intelligence Platform (Kalvium-First) – Product Requirements Document (PRD)

Role & Expectations

Act as a Principal Software Architect, Staff AI Engineer, Product Manager, UX Architect, and Technical Lead with 20+ years of experience building scalable EdTech, AI, and SaaS products.

Design and build a production-grade platform called **Placement Intelligence Platform**.

This is NOT a generic AI mock interviewer.

This is a continuously learning placement intelligence ecosystem that becomes smarter after every interview conducted by students.

The platform should be designed as:

- Kalvium-first
- Multi-tenant ready
- Extensible beyond Kalvium
- AI-native
- Data-driven
- Scalable
- Production-ready

Vision

Students repeatedly fail interviews because:

- They don't know what companies are currently asking.
- They don't know their actual weaknesses.
- They don't receive enough realistic interview practice.
- They lack personalized feedback.
- Interview experiences remain scattered and are lost over time.

The platform must solve this problem by combining:

1. Interview Experience Collection
2. Company Intelligence
3. AI Mock Interviews
4. Student Progress Tracking
5. Personalized Learning Recommendations
6. Placement Readiness Analytics

Core Product Philosophy

The moat is NOT AI.

The moat is:

- Real interview experiences
- Company-specific intelligence
- Historical placement data
- Student performance trends
- Continuously growing knowledge base

Every interview submitted should improve the platform.

User Roles

Student

Can:

- Login
- Upload resume
- View dashboard
- Submit interview experiences
- Explore company intelligence
- Take AI mock interviews
- View interview history
- View performance trends
- View recommendations

Admin

Can:

- Manage companies
- Manage interview experiences
- Moderate submissions
- Approve flagged content
- Monitor platform health
- View analytics

Future Role (Not MVP)

Mentor

Can:

- View student performance
- Analyze cohort weaknesses
- Track readiness metrics

Do not implement initially but design database for future support.

Module 1: Authentication

Requirements:

- Email/password authentication
- Google OAuth
- Role-based access control
- Secure session management

Student Profile Fields:

- Full Name
- Email
- Kalvium ID
- Batch
- Graduation Year
- Resume
- LinkedIn URL
- GitHub URL

Module 2: Student Dashboard

After login student should see:

Profile Summary

- Name
- Batch
- Resume status
- Target companies

Placement Statistics

- Interviews attempted
- AI interviews completed
- Contributions submitted
- Company reports viewed

Recent Activity

- Recent mock interviews
- Recent company reports
- Recent submissions

Performance Overview

Display:

- Communication
- Technical Depth
- Problem Solving
- Behavioral Skills
- Project Discussions

Trend graphs required.

Module 3: Interview Experience Collection

This is the most important module.

Purpose:

Create a continuously growing interview intelligence database.

Submission Form

Required Fields:

Company Name

Role

Interview Date

Round Type

Options:

- OA
- Technical
- HM
- HR

- System Design
- Other

Difficulty

Options:

- Easy
- Medium
- Hard

Outcome

Options:

- Selected
- Rejected
- Waiting
- Unknown

Questions Asked

Allow multiple entries.

For each question:

Topic

Examples:

- Arrays
- Binary Search
- SQL
- DBMS
- React
- OOP

Exact Question

Could Answer?

Options:

- Yes
- Partially
- No

Student Notes

Optional

AI Processing Layer

After submission:

AI should:

Extract:

- Topics
- Skills
- Technologies
- Interview Patterns

Generate structured metadata.

Store both:

Raw Data

+ AI Structured Data

Module 4: Company Intelligence Engine

Purpose:

Transform raw interview experiences into actionable insights.

Company Page Example:

Lowe's

Display:

Total Experiences

Interview Rounds

Most Asked Topics

Frequently Asked Questions

Difficulty Trends

Recent Experiences

Success Patterns

Common Weaknesses

Questions by Round

Technical

HM

HR

OA

Time Sensitivity

Recent data is more important.

Weight:

Last 90 Days -> High Priority

Older Data -> Lower Priority

Module 5: Resume Intelligence

Student uploads resume.

AI extracts:

- Skills
- Projects
- Experience
- Technologies
- Domains

Store structured profile.

Generate:

Resume Insights

Missing Skills

Strength Areas

Potential Interview Topics

Module 6: AI Mock Interview System

Critical Feature.

Interview Types

Behavioral

Technical

Hiring Manager

Project Discussion

Company Specific

Custom

Difficulty Modes

Easy

Medium

Hard

Resume-Aware Interviews

Questions must be generated using:

Resume

+ Target Company

+ Past Interview Data

Example:

Student mentions FastAPI.

AI should ask:

Why FastAPI?

How does async work?

Difference between FastAPI and Flask?

Adaptive Follow-Ups

Mandatory.

The AI must generate follow-up questions dynamically.

Do NOT use fixed flows.

Follow-up questions must depend on previous answers.

Interview Modes

Text Mode

Voice Mode

Both supported.

Module 7: Voice Interview Engine

Student speaks.

Pipeline:

Voice Input

Speech-to-Text

Interview Processing

Evaluation

Feedback

Text-to-Speech Response

Recommended:

Whisper

Piper

or equivalent open-source alternatives.

Module 8: Evaluation Engine

After interview:

Generate:

Communication

Technical Depth

Confidence

Problem Solving

Behavioral Skills

Project Understanding

Do NOT generate arbitrary scores.

Every score must be explainable.

Example:

Communication: 7/10

Reason:

Clear explanations but frequent pauses.

Module 9: Interview Replay

Store:

Question

Student Answer

Evaluation

Feedback

Ideal Answer

Students should be able to review interviews later.

Module 10: Weakness Detection Engine

Track recurring weaknesses.

Example:

SQL appears weak across:

- Mock 1
- Mock 2
- Mock 3

System should detect:

Recurring Weakness:

SQL Joins

Recommended Actions:

- Practice SQL Sheet
- Take SQL Mock

Module 11: Learning Recommendations

Based on:

Resume

Interview History

Weaknesses

Target Companies

Generate:

Topics to Study

Priority Ranking

Practice Plans

Suggested Mock Interviews

Module 12: Placement Readiness Engine

Generate readiness score.

Must be explainable.

Example:

Lowe's Readiness

78%

Based On:

Technical

Communication

Behavioral

Project Discussions

Company-specific competencies

Never generate opaque scores.

Module 13: Knowledge Base Search

Students can search:

Company

Topic

Question

Technology

Example:

Search:

Binary Search

Returns:

Companies that asked it

Frequency

Difficulty

Related Questions

Data Model Requirements

Design normalized database schema.

Core Tables:

Users

Profiles

Companies

Interview Experiences

Interview Questions

Interview Outcomes

Company Analytics

Mock Interviews

Mock Interview Questions

Evaluations

Weaknesses

Recommendations

Resume Data

Activity Logs

Roles

Permissions

Design for future scalability.

AI Architecture

Use modular AI services.

Services:

Resume Parser

Interview Generator

Follow-Up Generator

Evaluator

Recommendation Engine

Knowledge Extractor

Analytics Engine

All services should be independently replaceable.

Technical Stack

Frontend

- Next.js
- TypeScript
- Tailwind
- Shadcn UI

Backend

- FastAPI

Database

- PostgreSQL

Vector Store

- pgvector

AI

- Gemini initially
- Ollama support later

Speech

- Whisper
- Piper

Authentication

- Supabase Auth

Deployment

Frontend -> Vercel

Backend -> Render / Railway

Database -> Supabase PostgreSQL

Non-Functional Requirements

Performance

- Fast page loads
- Async processing

Security

- JWT Authentication
- RBAC
- Data encryption

Scalability

- Multi-tenant ready
- Thousands of users
- Millions of interview records

Observability

- Logging
- Monitoring
- Error tracking

Maintainability

- Clean Architecture

- Domain-driven design
- Modular services

Success Metrics

Student Engagement

Interview Experience Contributions

AI Interview Completion Rate

Company Intelligence Coverage

Placement Readiness Accuracy

Student Improvement Trends

Knowledge Base Growth

Final Objective

Build a platform that becomes smarter after every interview experience submitted.

The long-term vision is to become the operating system for placement preparation by combining:

- Real interview intelligence
- Personalized AI coaching
- Company-specific preparation
- Progress tracking
- Readiness analytics

The platform should continuously learn from student experiences and provide increasingly accurate guidance to future students.